

EXERCISES

(1) Resolve the following into partial fractions :

~~(a)~~ $\frac{1}{x(x+1)}$

~~(b)~~ $\frac{5x+1}{(x-1)(x+2)}$

~~(c)~~ $\frac{3x+2}{(x+1)(x-3)}$

~~(d)~~ $\frac{x^2}{(x+1)(x+2)(x+3)}$

(2) Resolve the following into partial fractions :

(a) $\frac{1}{x^2-1}$

(b) $\frac{1}{1-x^2}$

(c) $\frac{x-2}{x^2-x}$

~~(d)~~ $\frac{x^2+1}{x(x^2-1)}$

(e) $\frac{x+4}{x^2+x}$

~~(f)~~ $\frac{5x+7}{x^2+2x-3}$

~~(g)~~ $\frac{x+3}{x^3+7x^2+10x}$

~~(h)~~ $\frac{x^2-2x+3}{x^3+x}$

(i) $\frac{x-1}{x(x^2+1)}$

~~(j)~~ $\frac{x^2-x+3}{(x-2)(x^2+1)}$

(k) $\frac{x-2}{x^3+1}$

(l) $\frac{x^2+1}{x^3+x^2+x}$

(3) Resolve the following into partial fractions :

~~(a)~~ $\frac{1}{(x-1)^2(x+4)}$

~~(b)~~ $\frac{x+1}{x^3-x^2}$

(c) $\frac{3x-2}{x^3+2x^2}$

(4) Resolve the following into partial fractions :

(a) $\frac{x^3-1}{(x+1)(x+2)}$

(b) $\frac{x^2+7x-6}{(x-1)(2x-1)}$

(c) $\frac{x^3}{x^2-1}$

(d) $\frac{x^2-x+3}{x^2+x-6}$

(e) $\frac{x^2-2x+2}{x^2+2x-3}$

(f) $\frac{x^2+x+1}{x^2-1}$

(g) $\frac{x^3+x}{x^2-4}$

(h) $\frac{x^3-1}{x^3+1}$

~~(i)~~ $\frac{x^4}{x^2-1}$

ANSWERS

(1) (a) $\frac{1}{x} - \frac{1}{x+1}$

~~(b)~~ $\frac{2}{x-1} + \frac{3}{x+2}$

~~(c)~~ $\frac{1}{4} \left[\frac{1}{x+1} + \frac{11}{x-3} \right]$

~~(d)~~ $\frac{1}{2} \left[\frac{1}{x+1} - \frac{8}{x+2} + \frac{9}{x+3} \right]$

(2) (a) $\frac{1}{2} \left[\frac{1}{x-1} - \frac{1}{x+1} \right]$

(b) $\frac{1}{2} \left[\frac{1}{1-x} + \frac{1}{1+x} \right]$

(c) $\frac{2}{x} - \frac{1}{x-1}$

~~(d)~~ $\frac{1}{x+1} + \frac{1}{x-1} - \frac{1}{x}$

(e) $\frac{4}{x} - \frac{3}{x+1}$

~~(f)~~ $\frac{2}{x+3} + \frac{3}{x-1}$

~~(g)~~ $\frac{3}{10x} - \frac{1}{6(x+2)} - \frac{2}{15(x+5)}$

~~(h)~~ $\frac{3}{x} - \frac{2x+2}{x^2+1}$

(i) $\frac{x+1}{x^2+1} - \frac{1}{x}$

~~(j)~~ $\frac{1}{x-2} - \frac{1}{x^2+1}$

(k) $\frac{x-1}{x^2-x+1} - \frac{1}{x+1}$

(l) $\frac{1}{x} - \frac{1}{x^2+x+1}$

(3) ~~(a)~~ $\frac{1}{25} \left[\frac{1}{x+4} - \frac{1}{x-1} + \frac{5}{(x-1)^2} \right]$

~~(b)~~ $\frac{2}{x-1} - \frac{2}{x} - \frac{1}{x^2}$

(c) $\frac{2}{x} - \frac{1}{x^2} - \frac{2}{x+2}$

(4) (a) $x-3 + \frac{9}{x+2} - \frac{2}{x+1}$

(b) $\frac{1}{2} + \frac{2}{x-1} + \frac{9}{2(2x-1)}$

(c) $x + \frac{1}{2} \left[\frac{1}{x-1} + \frac{1}{x+1} \right]$

(d) $1 - \frac{3}{x+3} + \frac{1}{x-2}$

(e) $1 + \frac{1}{4} \left[\frac{1}{x-1} - \frac{17}{x+3} \right]$

(f) $1 + \frac{1}{2} \left[\frac{3}{x-1} - \frac{1}{x+1} \right]$

(g) $x + \frac{5}{2} \left[\frac{1}{x-2} + \frac{1}{x+2} \right]$